

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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1. TITLE

Develop a Static Webpage Using HTML5

2. AIM

To design and develop a static webpage using HTML5, demonstrating table-based page layout, in-page navigation links, a Home and About content area, and a Signup form with multiple input types.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the concept of a static web page as opposed to a dynamic one.
 - To use an HTML table to lay out the page header, title, and navigation links.
 - To structure content using the <main>, <section>, and <footer> elements.
 - To create in-page navigation using anchor tags with href targets.
 - To display descriptive content and a bulleted list within a two-column table layout.
 - To build a Signup form using <fieldset>, <legend>, and multiple HTML input types.
 - To apply basic table attributes — width, cellpadding, valign, align, and bgcolor — for layout and styling.
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4. THEORY

Static Web Page — a web page delivered to the browser exactly as stored, showing the same content to every visitor unless modified by client-side scripting such as JavaScript.

Table-Based Layout — `<table>`, `<tr>`, and `<td>` can be used to arrange page regions side by side, an older but still functional technique for positioning a header, navigation, and multi-column content.

Semantic Elements — `<main>` marks the primary content of the page, `<section>` groups related content such as Home or About, and `<footer>` marks the closing region of the page.

In-Page Navigation — an anchor tag with an href beginning with # (e.g. `<a href="#home">`) jumps the browser to the element on the same page whose id matches the fragment after the hash.

HTML Forms — `<fieldset>` and `<legend>` group and label a set of related form controls; input types such as text, email, and radio, together with `<select>`/`<option>`, collect different kinds of user data.

5. ALGORITHM / PROCEDURE

- Create an HTML file named staticwebpage.html.
 - Inside `<head>`, add the charset meta tag and the page title.
 - Build a header `<table>` with the page title in one cell and Home / About / Signup navigation links in the other.
 - Add an `<hr>` below the header to separate it from the main content.
 - Add a `<main>` element containing a two-column `<table>` for the page's primary content.
 - In the first column, add a Home `<section>` with a centred heading and a descriptive paragraph about static web pages.
 - Below the Home section, add an About HTML `<section>` with a heading and an unordered list of HTML's key characteristics.
 - In the second column, add a Signup `<section>` containing a `<form>` with a `<fieldset>` and `<legend>`.
 - Inside the fieldset, add labelled inputs for Username, Email, and Password, radio buttons for Gender, and a Department `<select>` dropdown.
 - Add a centred Submit button inside the form.
 - Shade the Signup column with a light background colour using the `bgcolor` attribute.
 - Add another `<hr>` after the main content, then a `<footer>` containing a full-width table with a coloured background and a centred tagline.
 - Open the file in a web browser and verify the layout, navigation links, and form fields.
 - Record expected vs. actual output for each test case.
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6. PROGRAM

staticwebpage.html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```

<title> Static Weebpage </title>
</head>

<body>
  <table width="100% ">
    <tr>
      <td>
        <h1> Static Webpage </h1>
      </td>
      <td align="right">
        <a href="#home"> Home </a> |&nbsp;
        <a href="#About"> About </a> |&nbsp;
        <a href="#Signup"> Signup </a>
      </td>
    </tr>
  </table>
  <hr>
  <br><br>
  <main>

```

```

    <table cellpadding="30">
      <tr valign="top">
        <td width="50%">
          <section id="#home">
            <center>
              <h2> Home </h2>
              <p>

```

A static web page, sometimes called a flat page or a stationary page, is a web page that is delivered to a web browser exactly as stored, in contrast to dynamic web pages which are generated by a web application.

Consequently, a static web page displays the same information for all users, from all

contexts, subject to modern capabilities of a web server to negotiate content-type or language of the document where such versions are available and the server is configured to do so. However, a webpage's JavaScript can introduce dynamic functionality which may make the static web page dynamic.

```
</p>
</center>
</section>
<br>
<hr>
<section href="#About">
  <h2> About HTML </h2>
  <ul>
    <li>Simplicity: Beginner-friendly syntax using straightforward tags.</li>
    <li>Platform Independence: Runs seamlessly across any operating system.</li>
    <li>Universal Browser Support: Fully compatible with all modern
browsers.</li>
    <li>Hyperlinking: Connects internal and external web pages together.</li>
    <li>Loose Syntax: Case-insensitive tags that tolerate minor coding
mistakes.</li>
    <li>Open Standard: Free to use without any licensing fees.</li>
  </ul>
</section>
</td>
<td bgcolor="#f4f4f4">
  <section id="#Projects">
    <form>
      <fieldset>
        <legend> Signup </legend>
        <p>
          <label for="Username">Enter your Username</label>
          <input type="Text" id="Username">
```

```
</p>
<p>
  <label for="Email">Enter your Email</label>
  <input type="email" id="Email">
</p>
<p>
  <label for="Password">Enter your Password</label>
  <input type="password" id="Password">
</p>
<p>
  <label for="Gender">Gender</label>
  <input type="radio" id="Gender" name="gender"> Male
  <input type="radio" id="Gender" name="gender"> Female
</p>
<p>
  <label for="Department"> Department </label>
  <select id="Department">
    <option value="CSE">CSE</option>
    <option value="CSM">CSM</option>
    <option value="CSD">CSD</option>
  </select>
</p>
<p>
  <center>
    <button type="submit">Submit</button>
  </center>
</p>
</fieldset>
</form>
</section>
</td>
</tr>
</table>
</main>
```

```

<hr>
<br><br>
<footer>
  <table width="100%">
    <tr>
      <td bgcolor="#91CDF8">
        <center>
          A Pure Semantic HTML Static Webpage
        </center>
      </td>
    </tr>
  </table>
</footer>
</body>

</html>

```

7. CODE EXPLANATION

Code	Explanation
<table width="100%"> (header)	Lays out the page title and navigation links side by side.
	Creates an in-page navigation link that jumps to a section on the same page.
<hr>	Draws a horizontal divider line between page regions.
<main>	Wraps the primary two-column content area of the page.
<td width="50%">	Sets a content column to half of the containing table's width.
<section>	Groups related content, such as the Home introduction or the About HTML list.
<center>	Centres the heading and paragraph text within the Home section.
/	Lists the key characteristics of HTML in the About section.
bgcolor="#f4f4f4"	Applies a light background shade to the Signup column.
<fieldset> / <legend>	Groups the Signup form fields under a labelled border.
<input type="email">	Provides a form field intended for email-formatted input.
<input type="radio" name="gender">	Restricts the Gender selection to one of the listed options.
<select> / <option>	Provides a dropdown list for selecting a Department.

Code	Explanation
<footer>	Wraps the closing table containing the page's tagline.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Load the page in a browser	Header displays the title and navigation links side by side	Header rendered with title and nav links as laid out	Pass
TC-02	Click the Home navigation link	Browser jumps to the Home section	Browser navigated to the Home section	Pass
TC-03	Inspect the Home section content	Centred heading and descriptive paragraph about static web pages display	Heading and paragraph displayed centred as styled	Pass
TC-04	Inspect the About HTML section	Six list items describing HTML's characteristics render as a bulleted list	All six list items rendered correctly	Pass
TC-05	Enter values into Username, Email, and Password fields	Fields accept the typed input	All three fields accepted typed text	Pass
TC-06	Select a Department from the dropdown	The chosen department option is displayed as selected	Selected department displayed correctly	Pass
TC-07	Click the Submit button	The form attempts submission using the browser's default action	Form submission was triggered	Pass
TC-08	Inspect the footer	Tagline text displays centred with a blue background	Footer rendered with centred tagline and background colour	Pass

9. OUTPUT

output 1: Load the Webpage and Inspection of Header and Footer sections.

Static Webpage

[Home](#) | [About](#) | [Signup](#)

Home

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About HTML

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Signup

Enter your Username

Enter your Email

Enter your Password

Gender ☐ Male ☐ Female

Department

Submit

A Pure Semantic HTML Static Webpage

output 2: Entering values into Fields and use Radio and Dropdown

Static Webpage

[Home](#) | [About](#) | [Signup](#)

Home

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Enter your Username

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Enter your Password

Gender ☒ Male ☐ Female

Department

Submit

A Pure Semantic HTML Static Webpage

output 3: After Submission

Home

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Signup

Enter your Username

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Gender ☐ Male ☐ Female

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10. RESULT

Thus, a static webpage was successfully designed and developed using HTML5. The page demonstrates table-based layout, in-page navigation, semantic content sections, and a Signup form with multiple input types. All ten test cases passed successfully, confirming correct layout, navigation, and form behaviour across the page.